

Theory of Computation

What is Pumping Lemma?

The word “Pumping Lemma” Means –

- **Lemma:** in mathematics refers to a mini theorem or refers to the intermediate theorem in a proof.
- **Pumping:** The word pumping refers to generating many input strings by pushing a symbol in an input string repeatedly.

There are **two Pumping Lemmas**, that are defined for -

- ✓ Regular Languages
- ✓ Context-Free Languages

Let's now learn about the Pumping Lemma in Theory of Computation in-depth.

What is pumping lemma in CFL?

The Pumping Lemma for Context-Free Languages (CFL) is a tool in formal language theory that helps prove the non-context-freeness of certain languages by demonstrating they cannot satisfy pumping properties.

Why we use Pumping Lemma?

It is a **theorem or one of the testing methods** to check whether the language is CFL or not

Is it a positive or negative test?

It is a negative test

Pumping Lemma for Context-free Languages

The Bar-Hillel lemma, also referred to as the pumping lemma for context-free languages, is a generalization of the pumping lemma for regular languages in formal language theory and computer science. It creates a characteristic that all context-free languages share.

The pumping lemma can be used to show by contradiction that a particular language is not context-free. In contrast, there are other conditions that must be met in order to guarantee that a language is context-free, such as Ogden's lemma or the Interchange lemma.

Implementation of Pumping lemma for Context-free Language

Pumping lemma for context-free languages If A is a context-free language, then there is a number p (the pumping length) where, if s is any string in A of length at least p , then s may be divided into five pieces $s = uvxyz$ satisfying the conditions

1. for each $i \geq 0$, $uv^i xy^i z \in A$,
2. $|vy| > 0$, and
3. $|vxy| \leq p$.

When s is being divided into $uvxyz$, condition 2 says that either v or y is not the empty string. Otherwise, the theorem would be trivially true. Condition 3 states that the pieces v , x , and y together have length at most p . This technical condition sometimes is useful in proving that certain languages are not context free.

The steps to prove that the language is not a context free by using pumping lemma are explained below –

- Assume that L is context free.
- The pumping length is n .
- All strings longer than n can be pumped $|w| \geq n$.
- Now find a string ' w ' in L such that $|w| \geq n$.
- Divide w into $uvxyz$
- Show that $uv^k xy^k z \in L$ for some k
- Then, consider the ways that w can be divided into $uvxyz$.
- Show that none of these can satisfy all the 3 pumping conditions at same time.
- w cannot be pumped (contradiction).